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J ²

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Клиент		Сервер
ClientHello ● supported_versions ● [signature_algorithms]* ● [key_share]* ● [supported_groups]* ● pre_shared_key ● psk_key_exchange_modes	→	
		ServerHello supported_versions ● pre_shared_key ●
		EncryptedExtensions
		Finished
		Application Data*
	←	NewSessionTicket*
		KeyUpdate*
Finished	→	
Application Data	↔	Application Data
	←	NewSessionTicket*
KeyUpdate*	↔	KeyUpdate*

Используемая система обозначений:

●	– расширения, посылаемые в рамках сообщения, под которым они указаны;
*	– опциональные данные;
[]	– расширение, посылаемое клиентом в режиме возобновления соединения для обеспечения возможности перехода к полной схеме обмена сообщениями;
	– сообщения, защищенные на ключах, выработанных из секретного значения [*sender*]_*handshake*_*traffic*_*secret* (см. подробнее 8.4);
	– сообщения, защищенные на ключах, выработанных из секретного значения [*sender*]_*application*_*traffic*_*secret*_*N* (см. подробнее 8.4).

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Клиент		Сервер
ClientHello: ● supported_versions ● [signature_algorithms]* ● key_share ● supported_groups ● pre_shared_key ● psk_key_exchange_modes	→	
		ServerHello: supported_versions ● key_share ● pre_shared_key ●
		EncryptedExtensions: supported_groups* ●
		Finished
		Application Data*
	←	NewSessionTicket*
		KeyUpdate*
Finished	→	
Application Data	↔	Application Data
	←	NewSessionTicket*
KeyUpdate*	↔	KeyUpdate*

Используемая система обозначений:

●	– расширения, посылаемые в рамках сообщения, под которым они указаны;
*	– опциональные данные;
[]	– расширение, посылаемое клиентом в режиме возобновления соединения для обеспечения возможности перехода к полной схеме обмена сообщениями;
	– сообщения, защищенные на ключах, выработанных из секретного значения [*sender*]_*handshake*_*traffic*_*secret* (см. подробнее 8.4);
	– сообщения, защищенные на ключах, выработанных из секретного значения [*sender*]_*application*_*traffic*_*secret*_*N* (см. подробнее 8.4).

Jbkmghd ² Ko_fZ h[f_gZ khh[s_gbyfb \ H36. (&'+(b L36. (&'+(
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g Z l _ d m s b c f h f _ g l

N h j f b j h \ Z g b _ i h e y W L F N H W B O L I H W L P H k h h [s _ g b y 1 H Z 6 H V V L R Q

A g Z q _ g i b _ e y W L F N H W B O L I H W L P H Z 6 H V V L R Q 7 ^ L F N H W B O L I H W L P H Z 6 H V V L R Q _ \ u r Z l v k _ d m g ^
^ g _ c G m e _ \ a g Z q _ g ^ b Z _ g g h j l h e y h a g Z q Z _ l h l b d _ ^ l h e `_ g u l v k j Z a b m k i h e v a h b Z m g ^ Z e _ g
D e b _ g b u _ ^ h e `_ g u o j Z g b l b d _ l o h e v r _ ^ g _ c g _ a Z \ b k h f l a g Z q _ g b y W L F N H W B O L I H W L P H
m ^ Z e y l b d _ l j Z g v r _ h d h g q Z g m b k y l Z g h \ e _ g k g _ h j \ h j h k j h d Z b a g b k h h l \ _ l k l \ b e h d Z e v g h c
i h e b l b d l k c _ j \ _ j f h `_ l k q b l Z f Z d k b f Z e v g ^ u h f i m k l b f u k j f h d h `f b a g b d _ l a Z g Z q _ g f b _ g v r __
q _ f m d Z a Z g h \ i h e _ W L F N H W B O L I H W L P H

< k h h l \ _ l k l \ b o > b o @ g Z k l h y s b j f _ b d h f _ g ^ Z p b ^ y h f i m k d Z _ k l h k a y ^ Z g g b h _ \ u b d _ l h \ j Z f d Z o
k h _ ^ b g _ g m b k y l Z g h \ e _ g k g h j h s v x L 3 6 . (& ' + b (L 3 6 . R (O b f h j Z [h l u j _ `b f h \ h k k l Z g h \
e _ g b k y h _ ^ b g _ g b Z d Z y h a f h `g h k i l h v a \ h e y g _ l h] j Z g b q _ i g h g ^ h e _ \ Z l k v j h d b a g b [s _] h d e x
q _ \ h] f h Z l _ j b Z e \ Z u j Z [h l Z g g h] j h Z f d Z b a g Z q Z e v g X h D H D Q G V K k D h N H b g _ g < b k y \ y a k w l b f
i j b j Z [h l _ i j h l h d h e \ Z k h h l \ _ l k l \ b o g Z k l h y s b j f _ b d h f _ g ^ Z p b a y g f Z q _ g i b _ e y 1 H Z 6 H V V L R Q 7 L F N
W L F N H W B O L I H W L P H g _ ^ h e `g h i j _ \ u r Z l v f b g b f m f Z b a k e _ ^ m x s b o a g Z q _ g b c
k _ d m g ^ ^ g _ c

j Z a g b p u f _ `^ m a g Z q _ g b y f b i _ j _ f _ g g u o J O R E D O B H [S L U D W L R Q B W L P H
d h e b q _ k l \ d Z m g j ^ h ^ b d l h \ Z g g j h m] b f e b h d Z e v g u h e b l b d Z k f b j \ _ j Z G Z i j b f _ j _ k e b
d f h f _ g l k h a ^ Z g l b y d _ l k Z _ j l b n b d Z e b _ g k Z h f h s v x d h l h j h j h k l Z g Z \ e b \ Z b e k h k h v ^ g k h h __ ^ b
g _ g b _ h l h a \ Z g l h k _ j \ _ j f h `_ l i j b g y l v j _ r _ g b _ h a Z i j _ l _ k h a ^ Z g b y g h \ u o l b
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? k e b d e b _ g h l h \ Z m l _ g l b n b p b j h \ Z h k e y h [f _ g Z P D L Q K D Q G V K D N H s N _ g b y f b k m s _
k l \ b l v S R V W K D Q Z m V K _ D N H h b d Z p b b x g ^ h e `_ g m d Z a Z Z v k r b j _ g b S R V W B K D Q G V K D N H B
k h h [s _ g k & b O L H Q W + k H e m q Z h e m q _ g ^ b Z y g g h j Z k r b j _ g b y l d e b _ g k Z j \ _ j f h `_ l h l j Z \ b l v
a Z i j h g Z Z m l _ g l b n b d Z p b b x g i l Z l _ f _ j _ k u e d k h h [s _ g k & y H U W L ¿ F D W k H 5 H T X \ H e v x W h c
f h f _ g l \ j _ f _ g b i h k e _ i h e m q _ g b y P D L Q K D Q G V K D N H k h h [s _ g b y) L Q L V K H G k

? k e b j Z k r b j _ g b S R V W B K D Q G V K D N H B n D o K Z W a Z g k h h [s _ g k & b O L H Q W + k H _ j O _ R g _
^ h e `_ g Z i j Z r b \ Z Z m l _ g l b n b d Z p b b x g i l Z k e a Z \ _ j r _ g b j y h l h d h e + Z D Q G V K D N H b \ g h f
k e m q Z _ d e b _ g l ^ h e `_ g h l \ _ l b l v h i h \ _ s _ g b _ f h [h r b [d _ X Q H [S H F W H G B P H V V

I j b i h e m q _ g & b H U W L ¿ F D W \ H 5 H T Z \ S H R V W K D Q Z m V K _ D N H b d Z p b b g ^ h e `_ g l \ _
l b l v k h h [s _ g b y Z b m l _ g l b n b d Z p b b x g ^ h e `_ g l \ _ l b l v k h h [s _ g b y & b H U W L ¿ F D W \ H & H U W L ¿ F D) W Q H 9 V K H G ? k e b d e b _ g l d e h g y _ l
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g b _ & H U W L ¿ F D W g k h ^ _ j `Z k v _ j l b n b d Z l h Z a f _ i h e y F H U W L ¿ F D W \ ^ H _ g u O v j Z W _ g m e x

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) L Q L V K H G g d p b 7 x U D Q V F U L S W Z + x D k V K _ P D L Q K D Q G V K D N H s N _ g b y _ j _ ^ Z g g u _ j Z f d Z o
^ Z g g h j k h _ ^ b g _ g b y h e v d l h _ S R V W K D Q G V K D N H D _ N g b y h l h j u h i j _ ^ _ e y x l _ d m s m j x h p _
^ m j m j h \ _ ^ _ g b Z y m l _ g l b n b d Z l p b k & H U W L ¿ F D W b H 5 H H T X H j V W D W j H b j h \ Z g b k b h h [s _ g b y
& H U W L ¿ F D b W k H H [H s U _ g l & y H U W L ¿ F D W & H H H T X H j & W W L ¿ F D _ W k H e 9 H g U [u e \ h i h k e Z g h
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k _ j \ _ j ^ h e `_ g u l v] h l h d i h e m q _ g j b x b a \ h e v g d h j e b q _ k l h Z h [s _ g b c ^ m f h f _ g l h f h l j Z \
d b k h h [s _ g k & y H U W L ¿ F D W b H 5 H _ T X H e v b W _ g u Z ^ Z g g h k _ h h [s _ g b D j h f _ l h] h d e b _ g h e m
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m d Z a Z g h b j y h q l h h l i j Z \ e y x s Z k y h j h g h Z [g h \ e y d e x q _ \ h Z l _ j b Z e j Z n b d Z e y n h j f b j h \ Z g b y
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> Z g g h k h h [s _ g b h `_ l [u l v h l i j Z \ e _ g k h h l \ _ l k l \ m x s k l h c j h g h \ o e x [h d h f _ g l \ j _ f _ g b
i h k e h l i j Z \ d b x P D L Q K D Q G V K D N H s N _ g b y Q L V K H G k h h [s _ g b H \ 8 S G D W H h l i j Z \ e _ g h
^ h P D L Q K D Q G V K D N H s N _ g b y Q L V K H G m q Z x s Z y k l h j h g ^ Z h e `_ g Z Z \ _ j r b l k v h _ ^ b g _ g k b i h
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h[t_^bg_gu \ h^bg ijb wlhf

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SODLQW?HQ'W'HFU\SWFH@VHTQGBQNHQFH DGGLWLRQDOBGDWD (1&UHF

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H i b k Z g g d Z e y x q _ \ Z b y j Z j o b i y _ ^ k l Z \ e _ g Z b ^ _ k o _ f u g Z j b k m g d _ ? _ h k g h \ g h j d b g p b i
a Z d e x q Z _ l k y e _ ^ m x s _ w f g l j h i b c g ^ u Z _ g g u h [h a g Z q _ g \ g e _ \ h c q Z k l k o _ f u g _ a Z \ b k y h l
k h h [s _ g b i c h l h d h e + Z D Q G V K i D _ N _ H k u e Z _ f u j p Z f d Z m k l Z g h \ e _ g b y m s _ k j h _ ^ b g _ g \ b l y h
\ j _ f y d Z k _ d j _ l g u a g Z q _ g f b y h ` _ k l \ 6 Z H F U g H Z W h ^ y s b _ k j y Z \ h c q Z k l k o _ f u a Z \ b k y h l k h
h [s _ g b i c h l h d h e + Z D Q G V K i D _ N _ H k u e Z _ f u j p Z f d Z m k l Z g h \ e _ g b y m s _ k j h _ ^ b g _ g b y w l h f m
f h] m l b k i h e v a h \ Z l v k y ^ e y \ u j Z [h l d b d e x q _ \ h] h f Z l _ j b Z e Z l j Z n b d Z

K _ d j _ l g u a g Z q _ g f b y h ` _ k l \ 6 Z H F U n H h W f b j m x l k k y i h f h s v x w g l j h i b c g ^ u Z _ g g u E C D H E
b P S K i h ^ Z x s b o k y Z \ o h ^ n m g d p b + y . f ') ([W U D F H W U L Y H 6 \ H k F h U H H \ W l k l \ b h k o _ f h c j _ ^
k l Z \ e _ g g h Z j b k m g d _ ? k e b h ^ g h b a a g Z q _ g 3 6 c . b e E C D H E g _ h i j _ ^ _ e _ y f h h f _ g l \ u j Z [h l d b
d e x q h Z [j Z [h l d b Z i b k _ b c i h ^ j Z a m f _ \ Z _ q k h y f _ k l h g _] b Z \ o h ^ n m g d p + b .) ([W U D F H W _ l k y
k l j h d Z + / H Q l j b w l h f h ^ g h \ j _ f _ g g h k m l k l \ b g l j h i b c g a u g Z q _ g P S K c b E C D H E g _ ^ h i m k d Z _ l k y
\ j Z f d Z o j Z [h l u i j h l 7 / d 6 e Z

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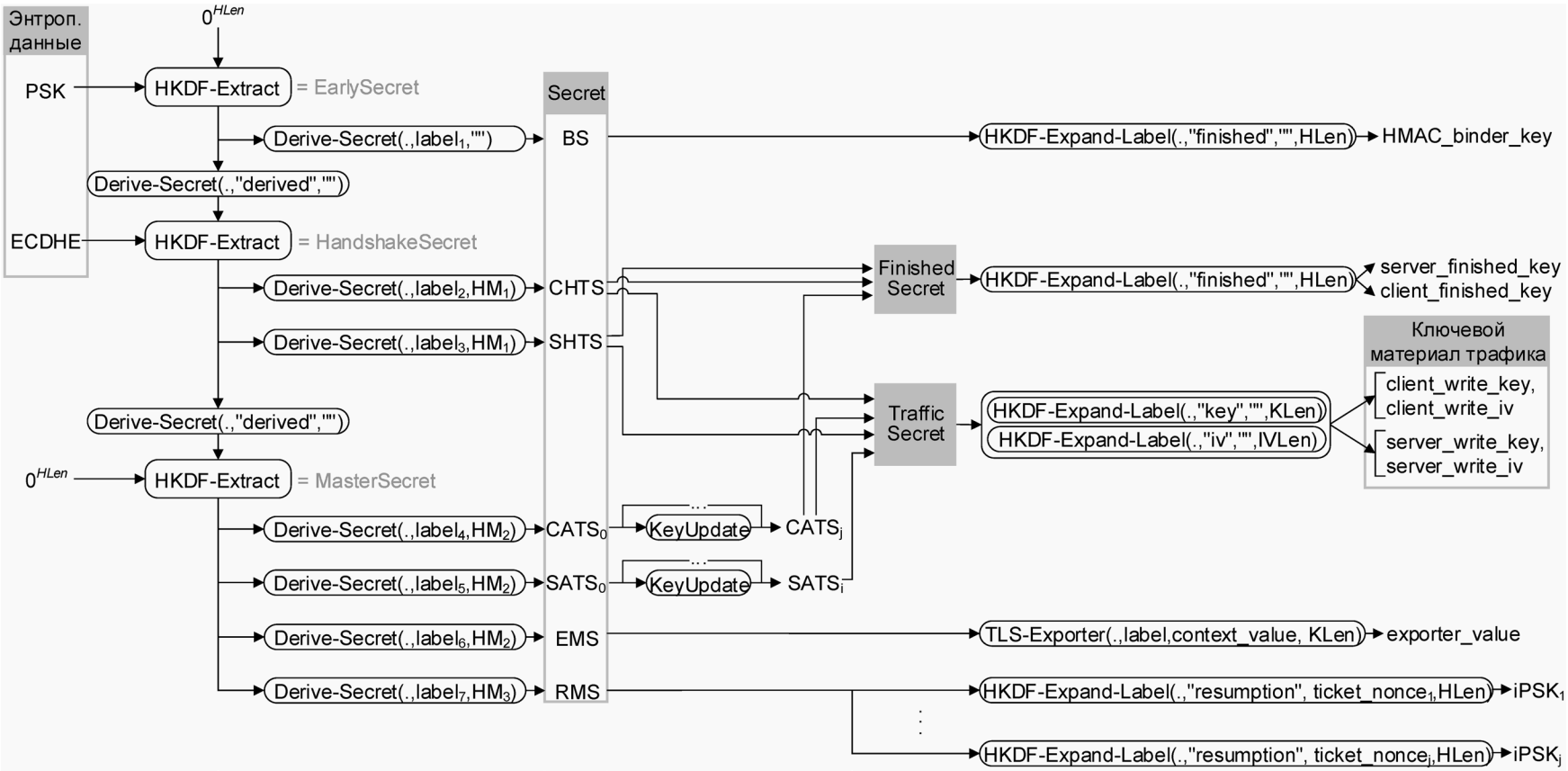

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ECDHE = VWU$_{FRRUGLQDW}$:ECDHE$_{HBOHQJWK}$

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